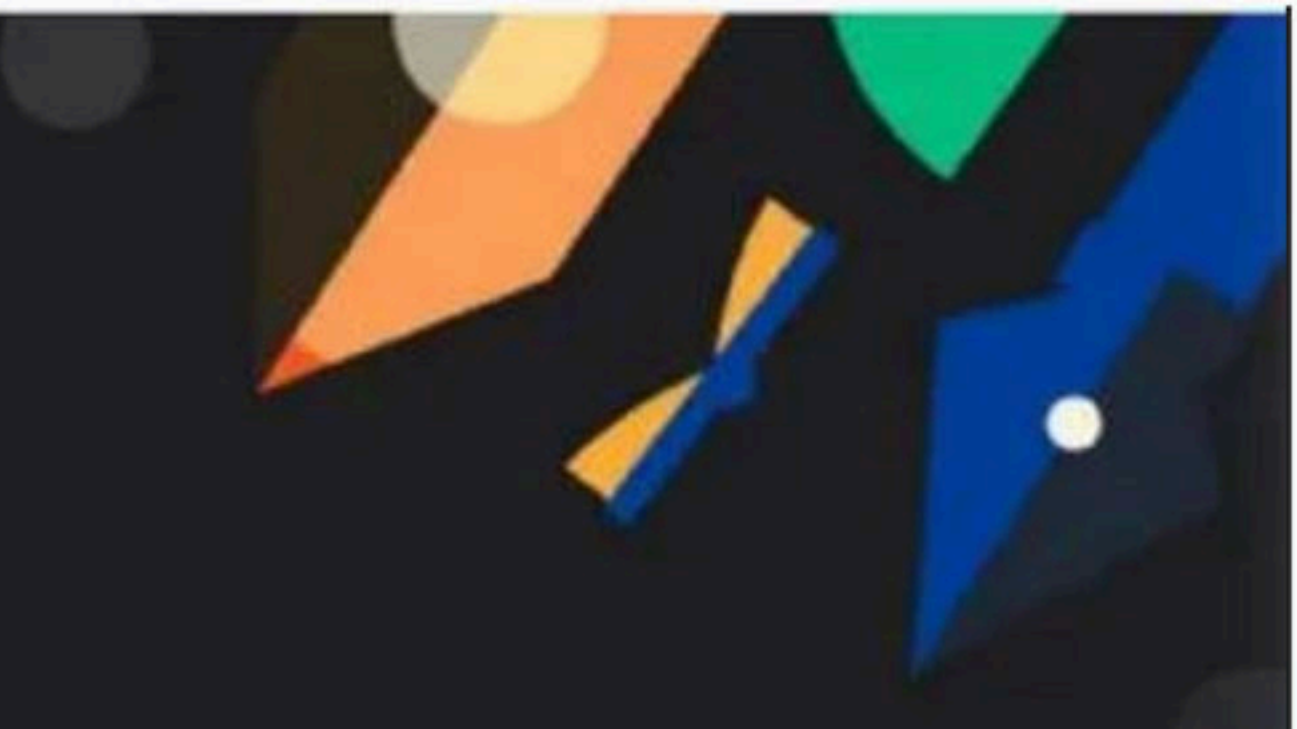




Graph Week-2 Doubt Class with Lakshay

Special class



Graph Class - 4

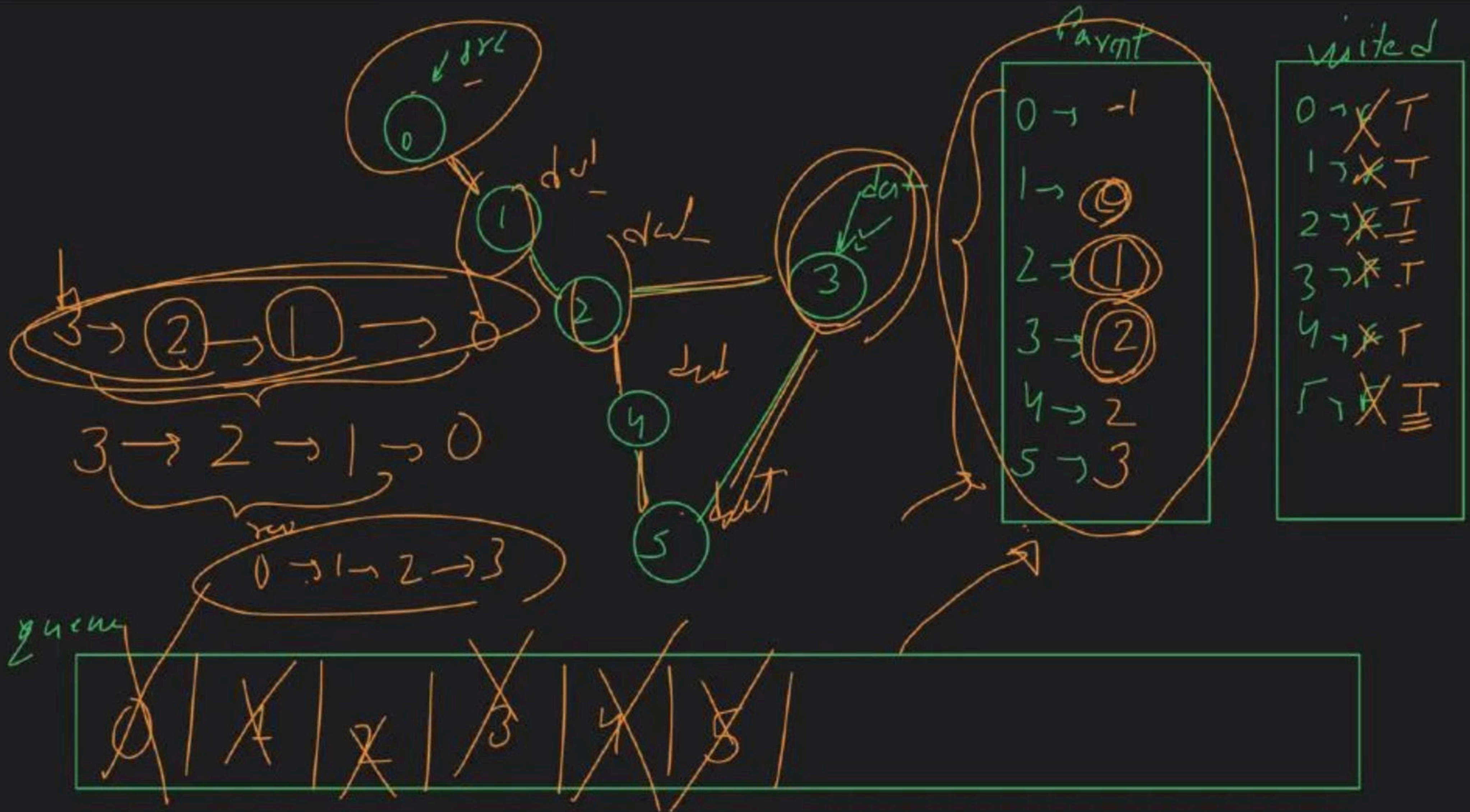
Special class

→ Shortest path

Observation - first time travel/unit → shortest path

① BFS
SSSP



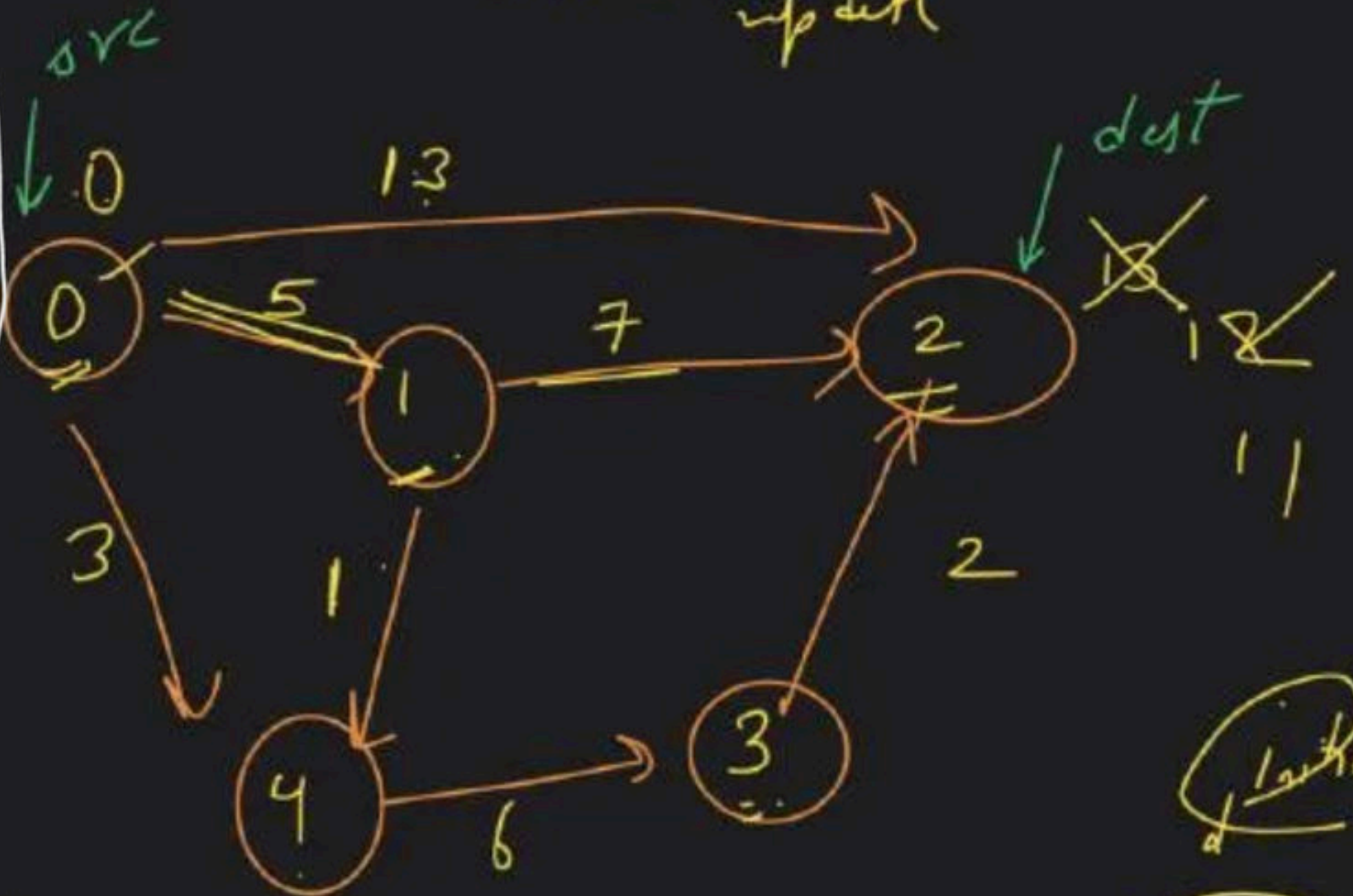
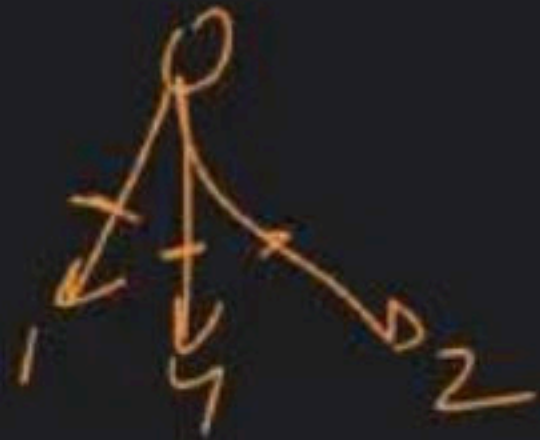


→ Directed graph → SSSP

[min distance]
 ↳ best shortest distance
 update

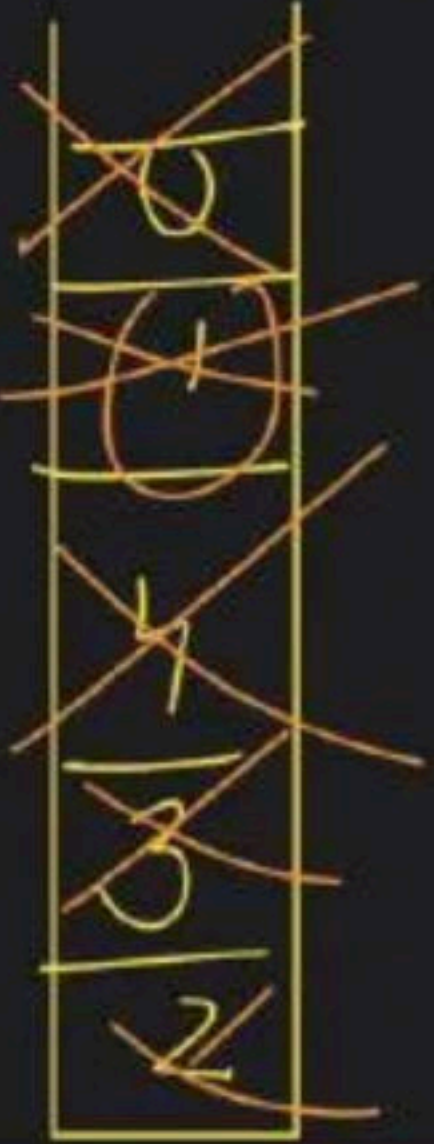
DFS

5 examples



dist
 $dist(src) = 0$

stack



0 1 2 3 4				
dist				
0	5	11	9	3

Kal

8:30 → 9:00

Graph Class - 5

Special class

Shortest path
==> ΔΔΔP → bfs
 → dfs

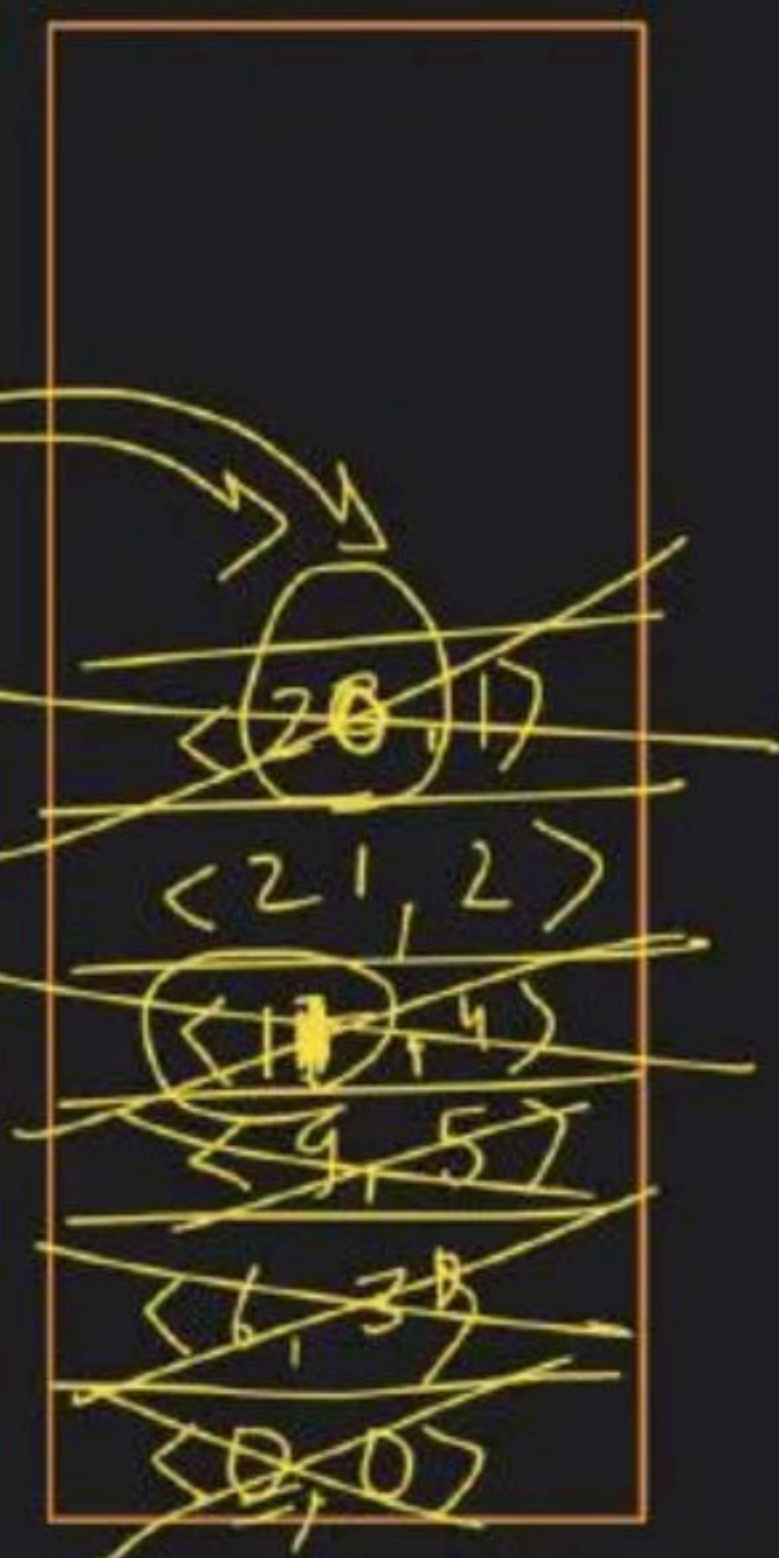
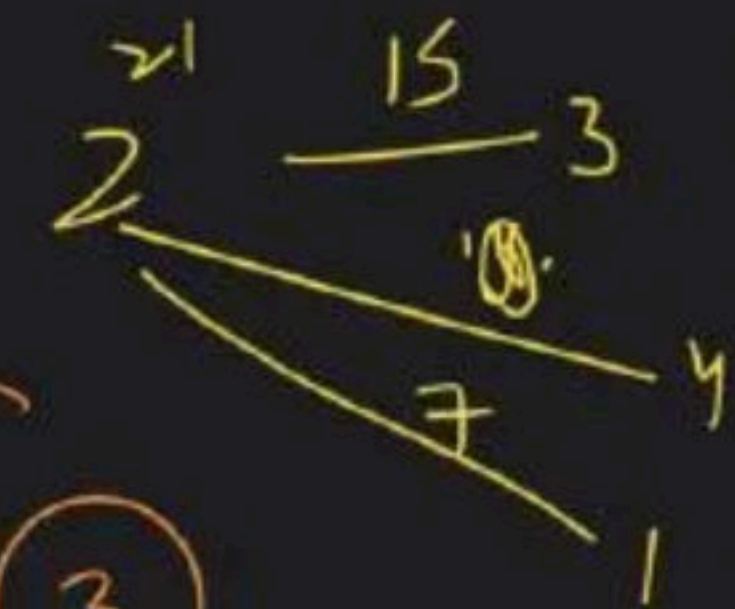
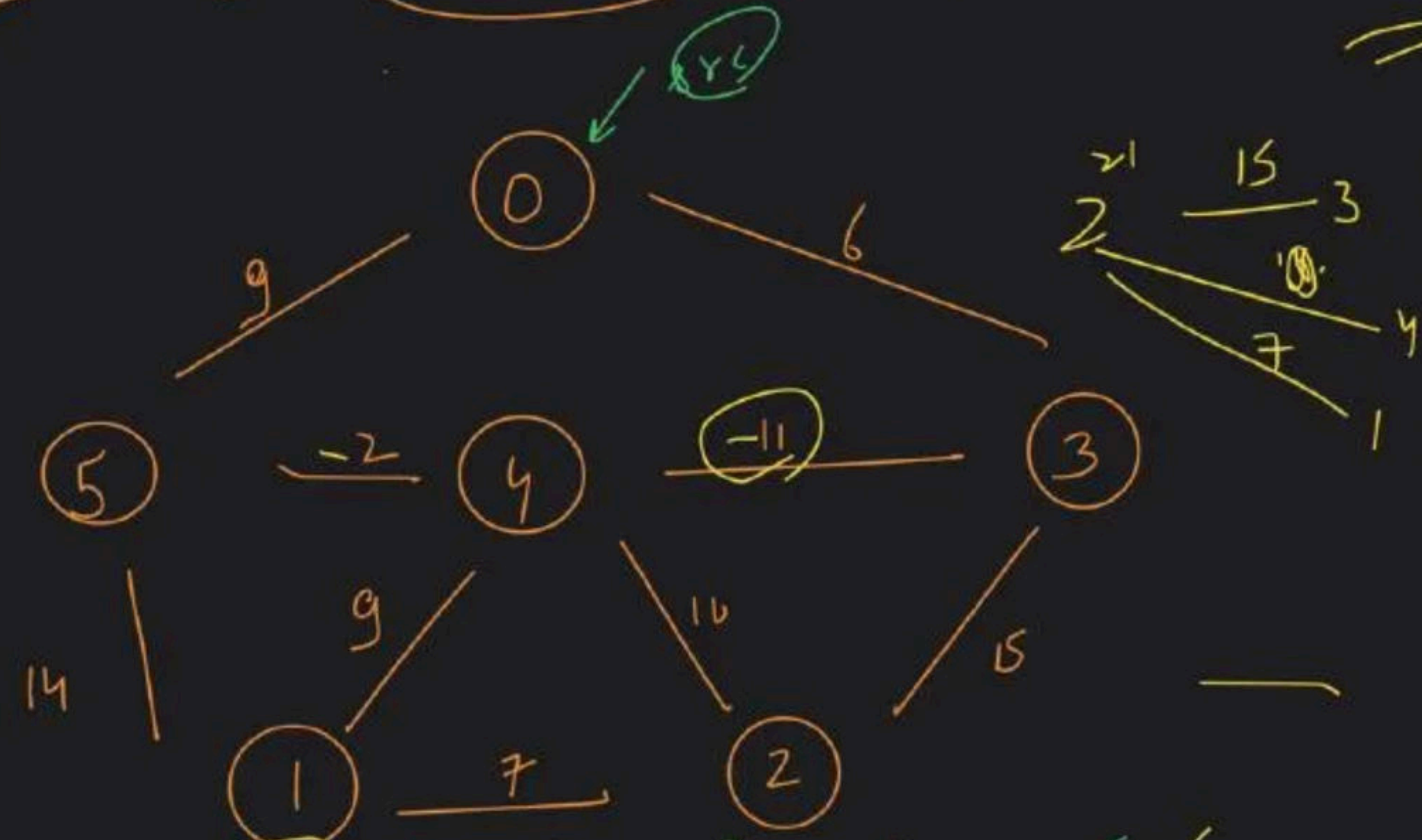
→ Dijkstra

SSSP

Algo: Greedy → P.Q
→ set

Set

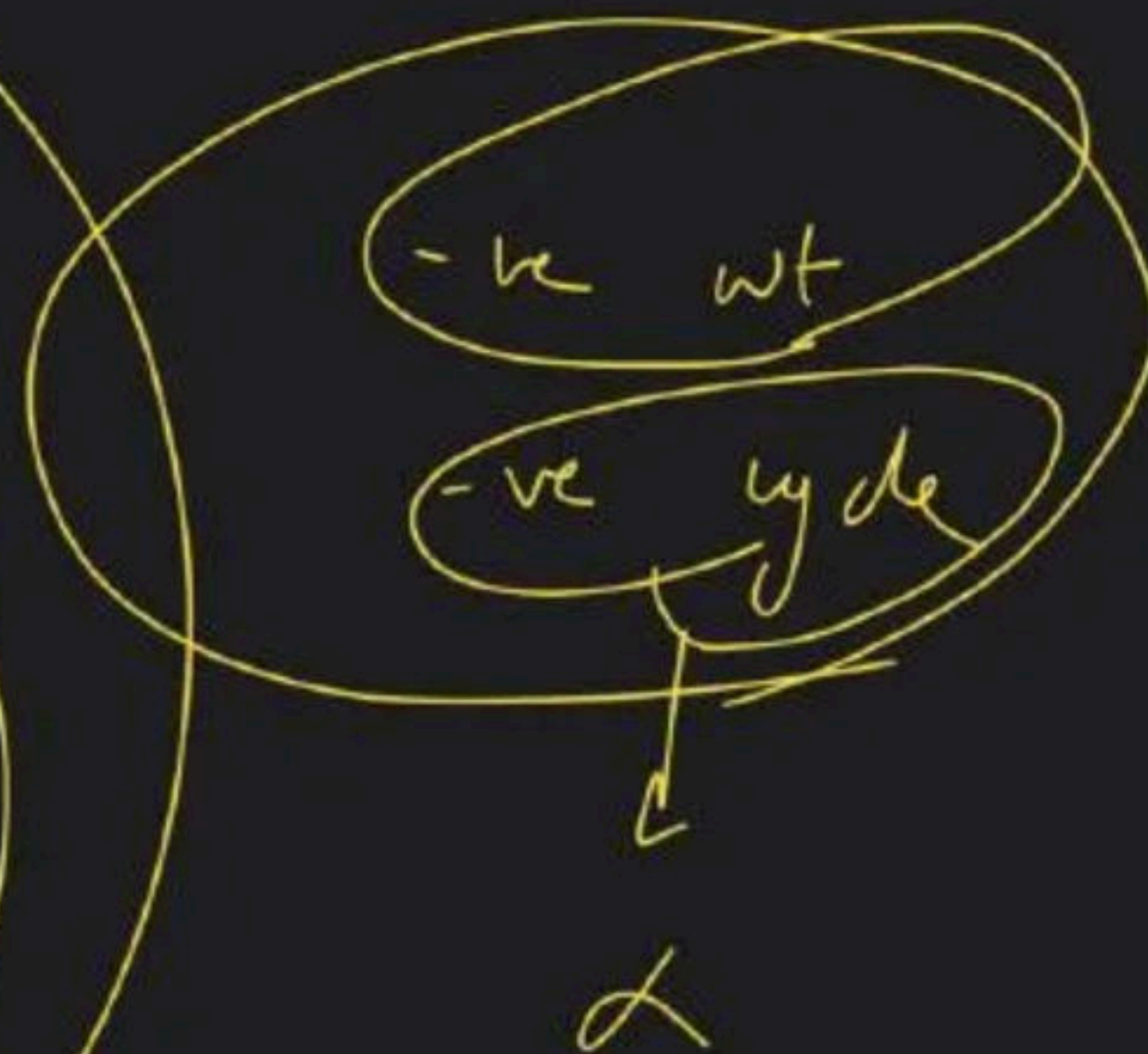
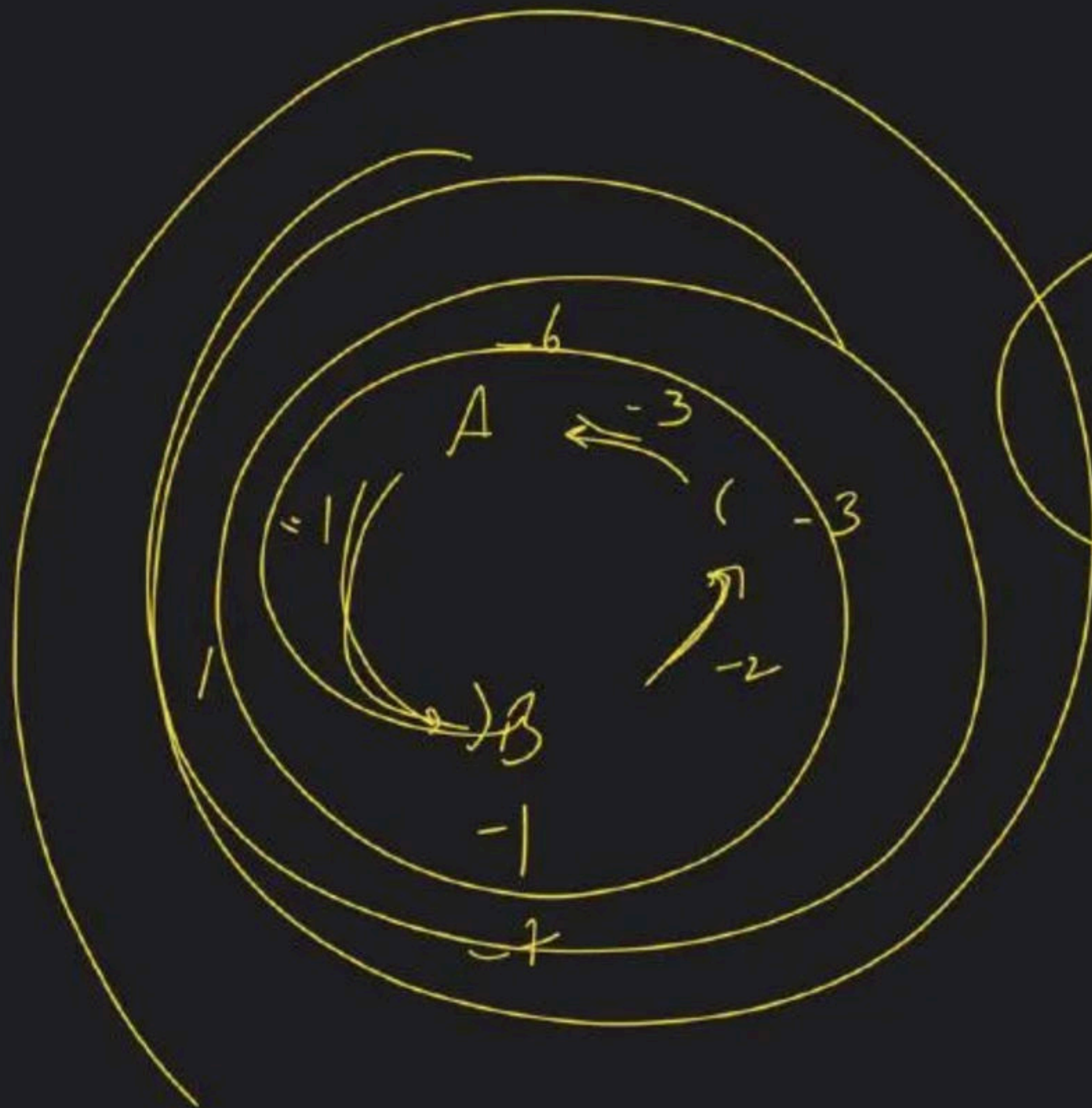
initial state
↓
 $dist[src] = 0$



pair <int, int>
↓ ↓
dist node

dist

0	1	2	3	4	5
0	20	21	6	11	9
0	20	21	6	11	9



→ Bellman Ford → 

Relaxation

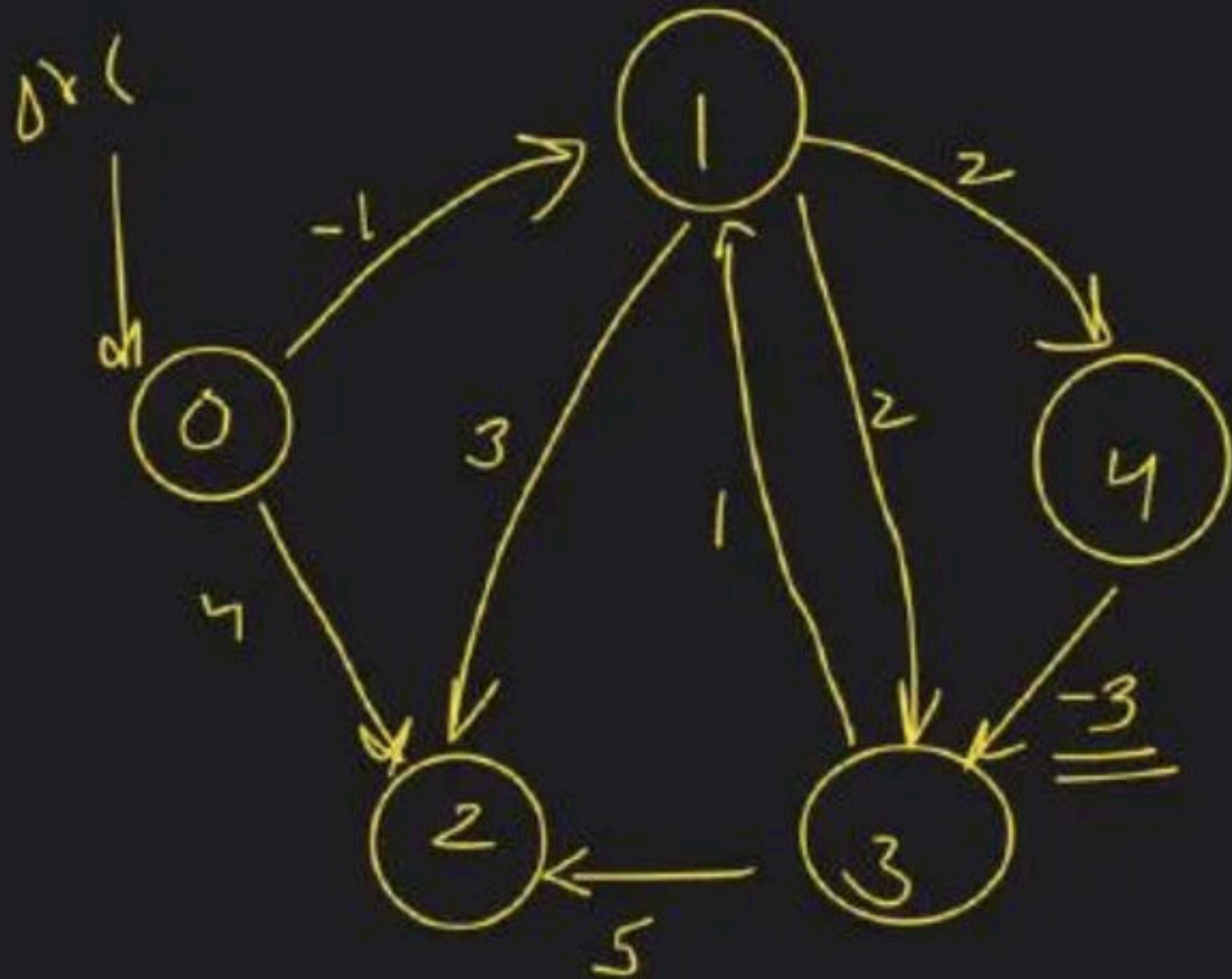
→ (N-1) times

why → ?

Result → shortest distance

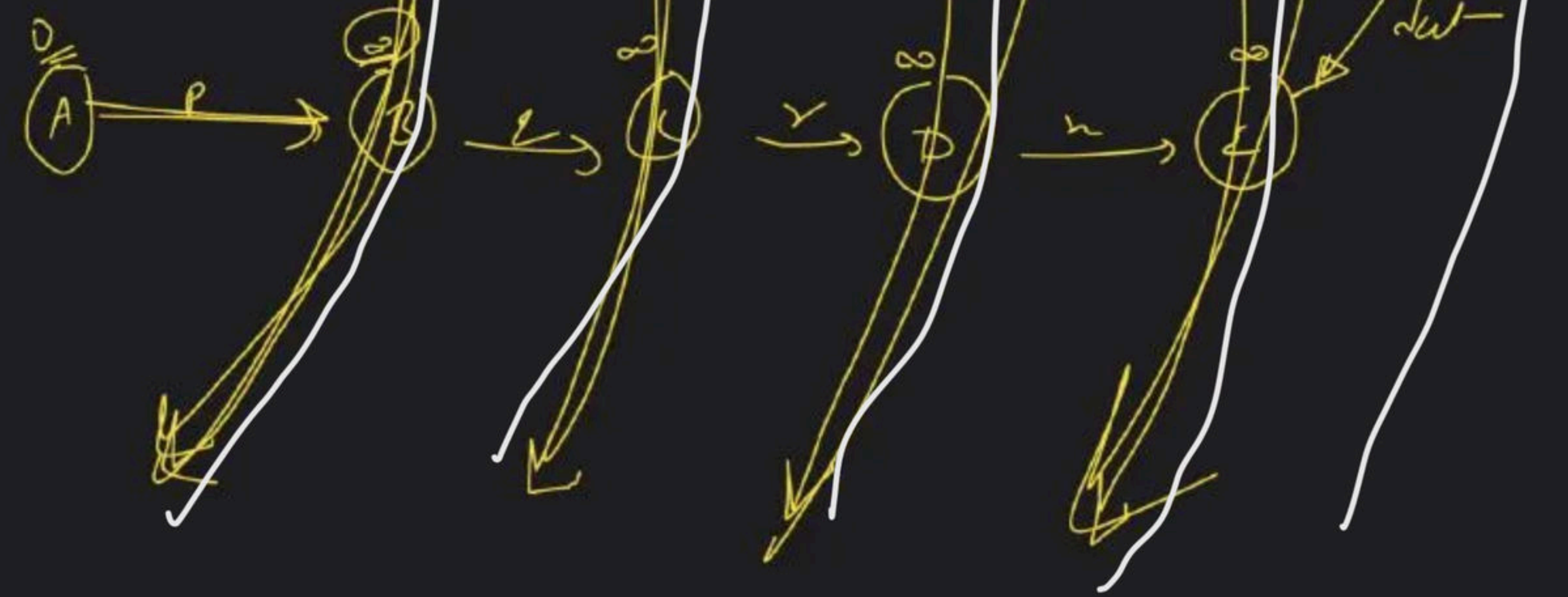
$$dist[u] + wt < dist[v]$$

graph

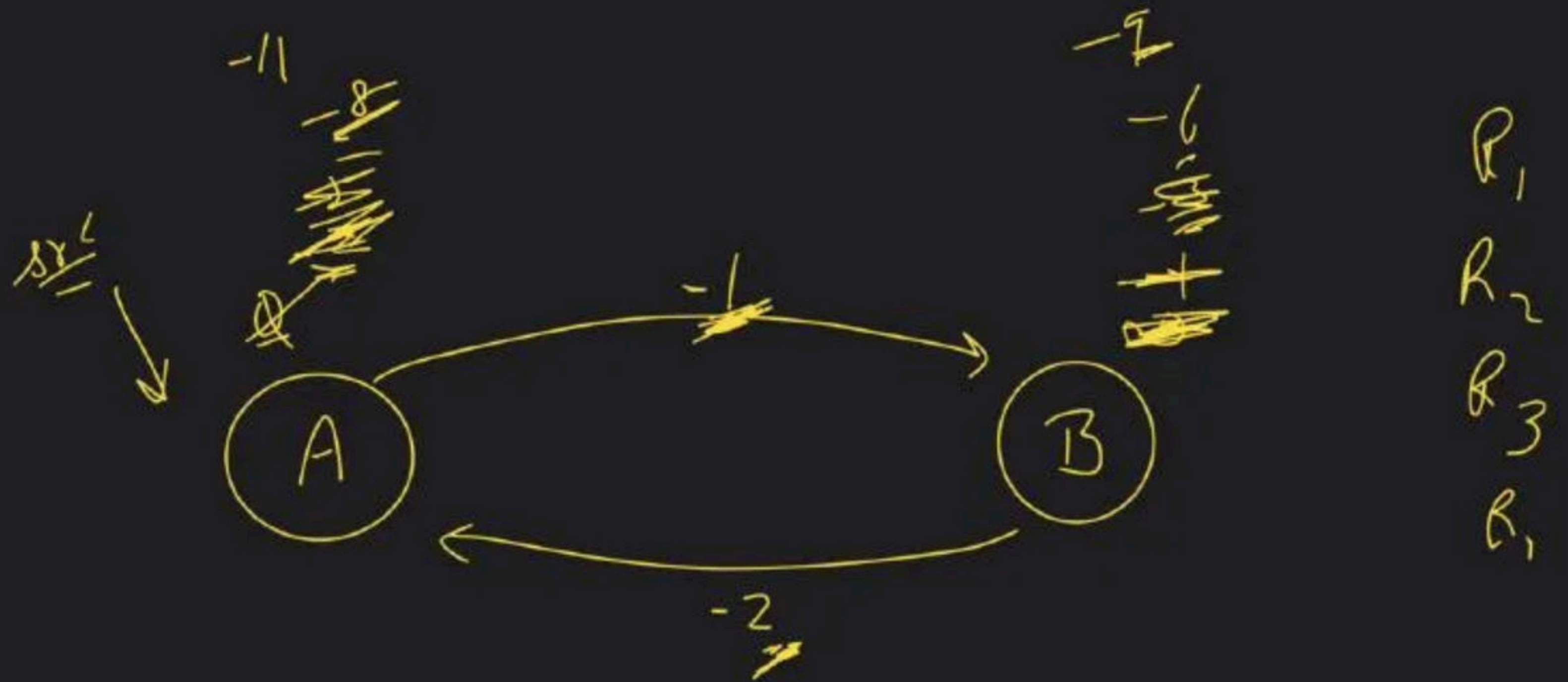




N=5



R₁



$-ve$ cycle $\rightarrow$ Relaxation $\rightarrow (N-1)$ time
 $\downarrow$
 update $\leftarrow$ True

Edge List

No. $V = 5$

0 $\xrightarrow{-1}$ 1

1 $\xrightarrow{2}$ 4

0 $\xrightarrow{4}$ 2

3 $\xrightarrow{5}$ 2

4 $\xrightarrow{-3}$ 3

1 $\xrightarrow{3}$ 2

1 $\xrightarrow{2}$ 3

3 $\xrightarrow{-1}$ 1

Relaxation #1

0	-1	2	-2	1
0	1	2	3	4

Relaxation #2

0	-1	2	-2	1
0	1	2	3	4

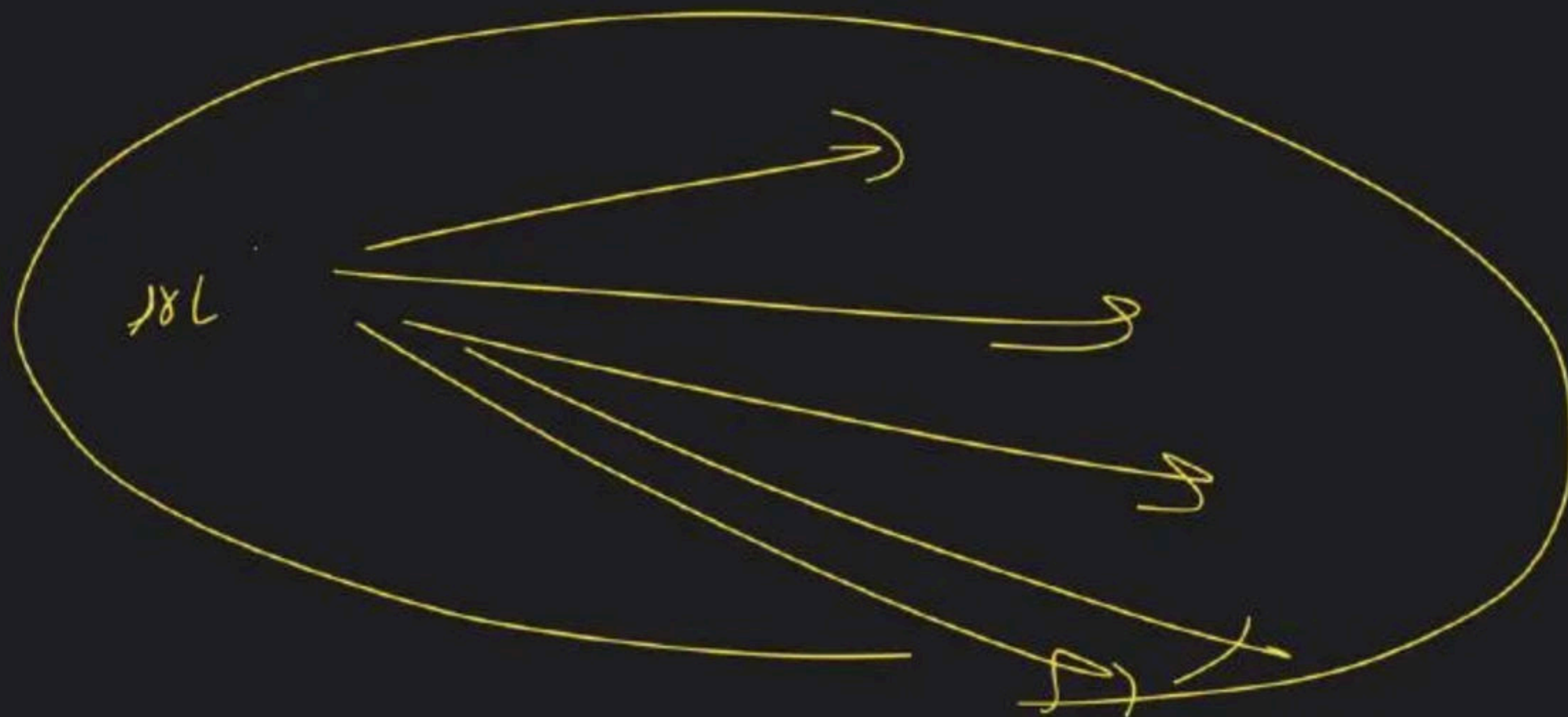
No change

#3 $\rightarrow$ No change

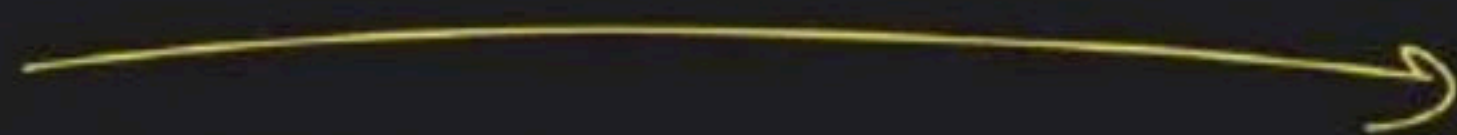
#4 $\rightarrow$ No change

break

SSSP

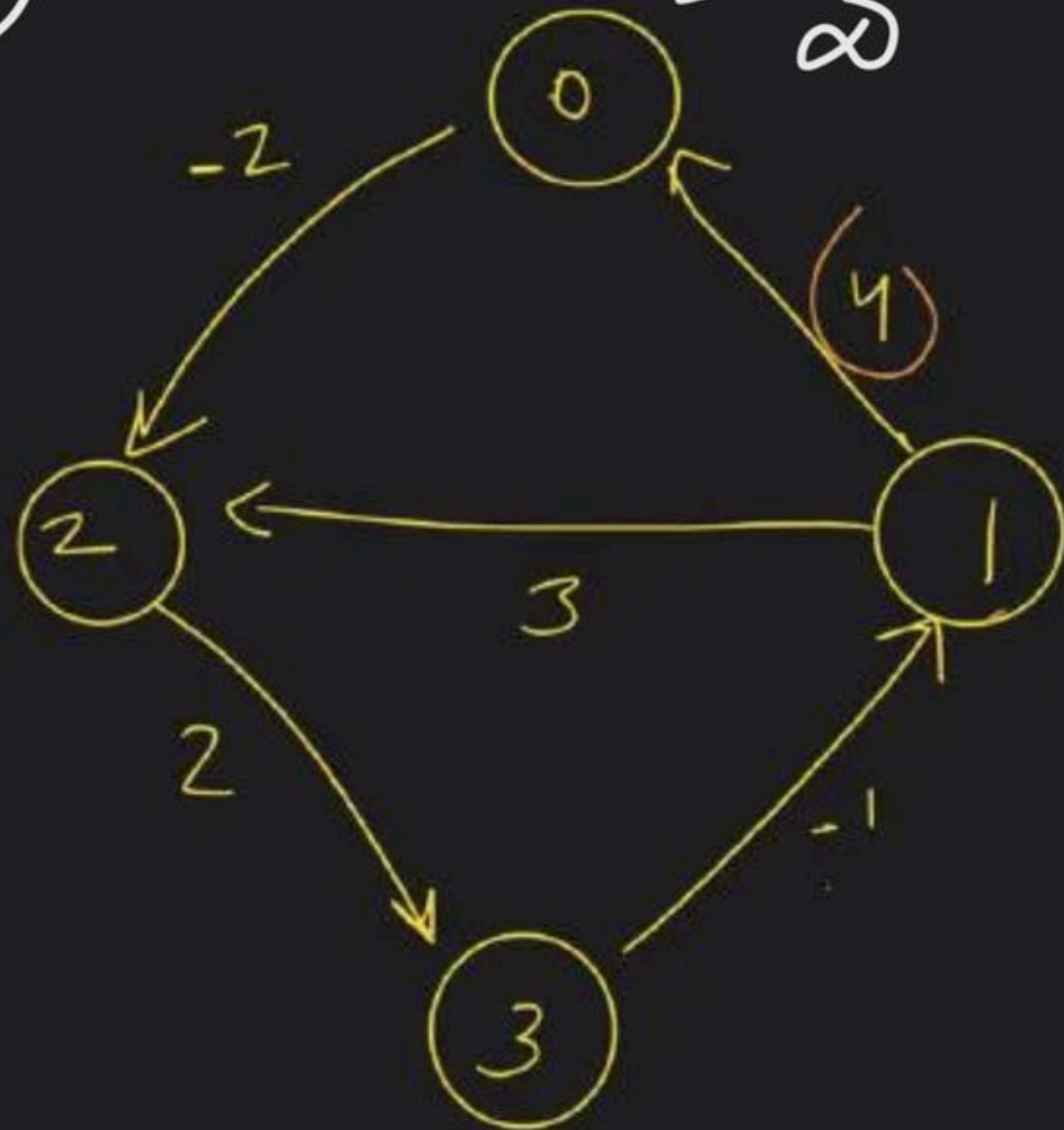


mvp



loyd
warshall
algo





(1) grab $\rightarrow$ dist $\rightarrow$ (adj)

(3) rat $\rightarrow$ ∞

$dist[u][v] = \min \rightarrow dist[u][v]$
 $dist[u][h] + dist[h][v]$

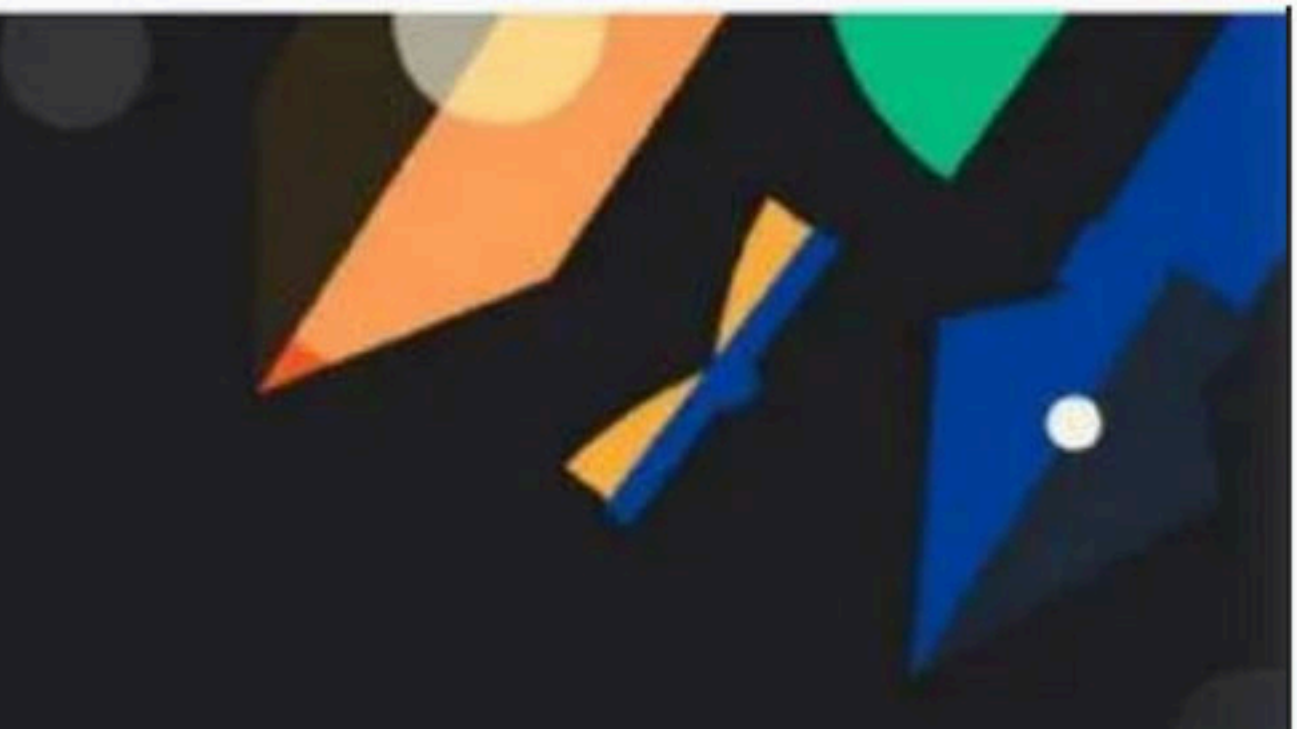
	0	1	2	3
0	0	-1	-2	0
1	4	0	3	4
2	5	1	0	2
3	3	-1	1	0

for (h $\rightarrow$ 0 $\rightarrow$ n)

{ for (v $\rightarrow$ 0 $\rightarrow$ n)

{ for (u $\rightarrow$ 0 $\rightarrow$ n)

log: $O(n^3)$



Graph Class - 6

Special class

④ Kosaraju Algo.

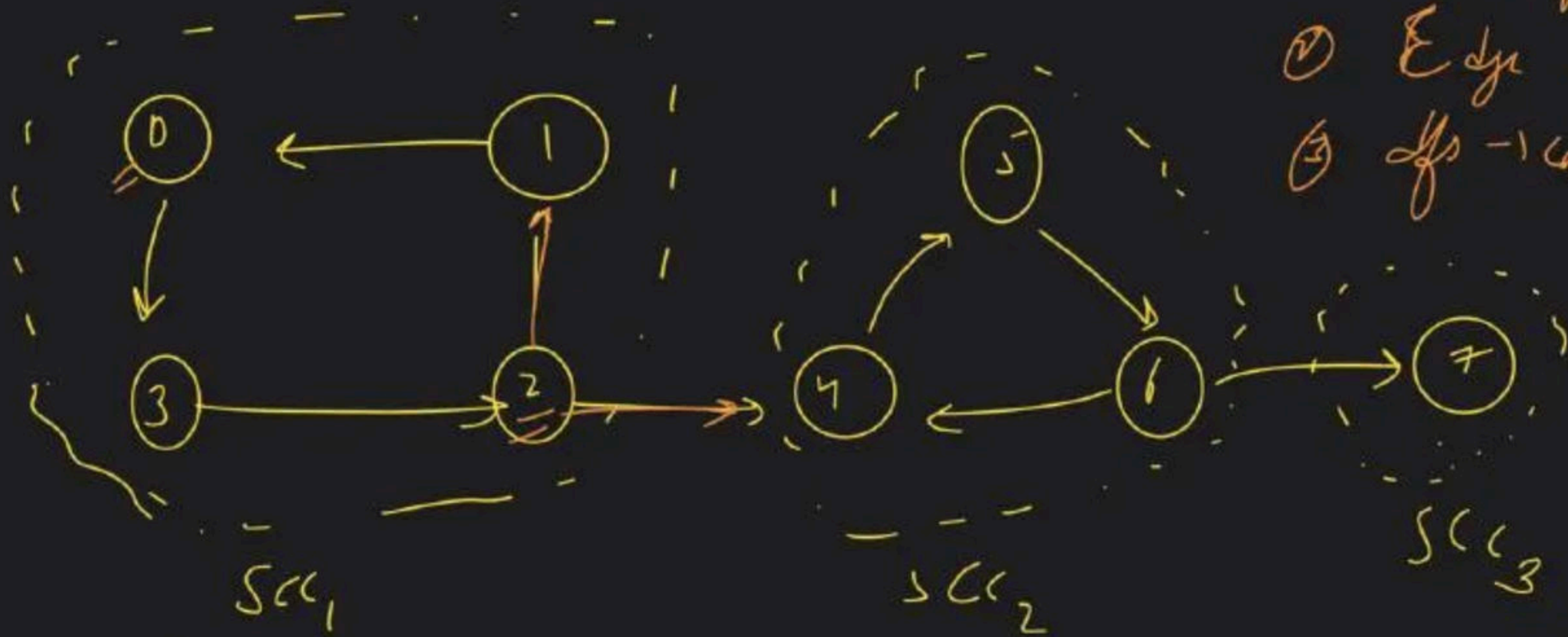


Strongly connected component



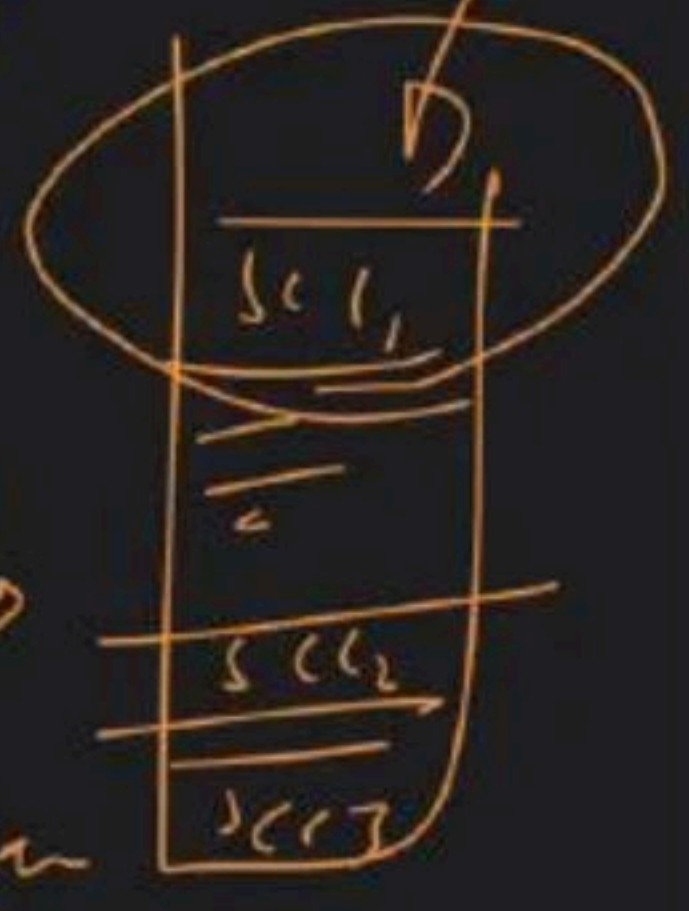
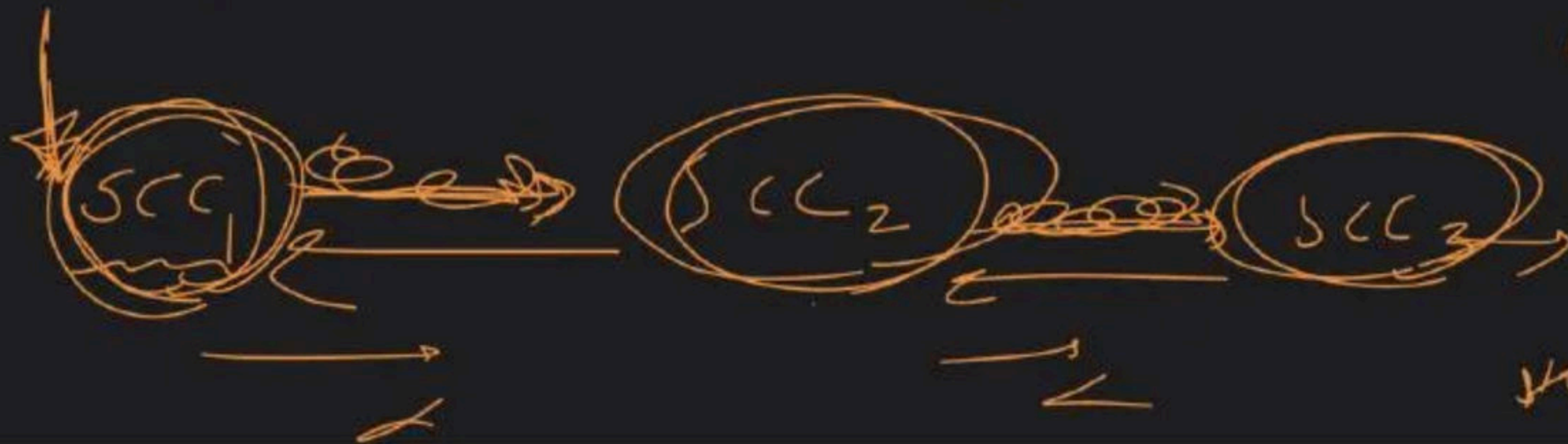
scc $\rightarrow$?

- ① Topological sorting
- ② Edge reverse
- ③ dfs - call - visit



SCC

Topo



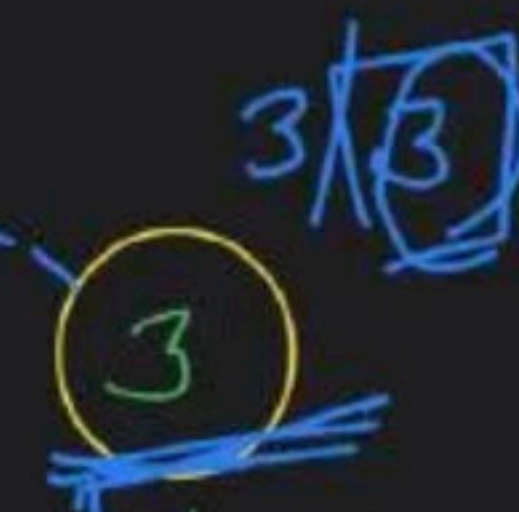
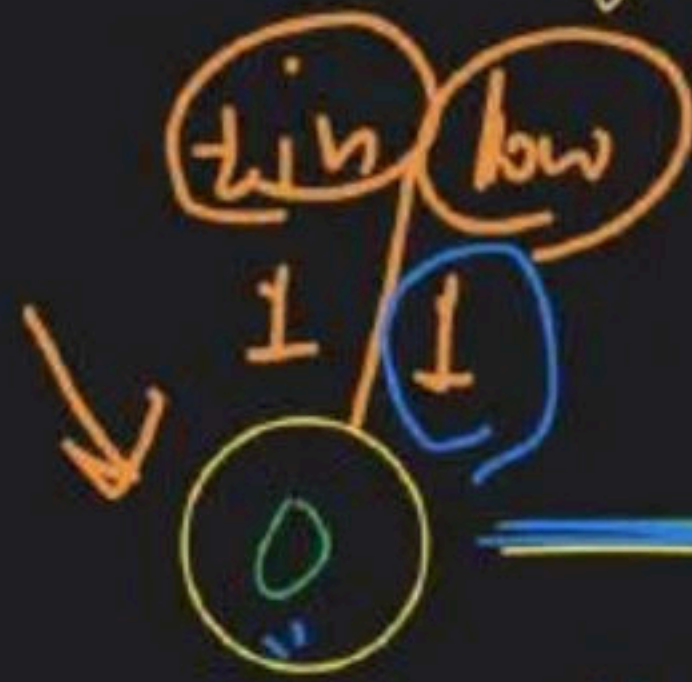
Bridge in a graph

Tarjan's

$$3 > 2$$

Bridge check

2 time



insertion time

tin already visited

low minimum time

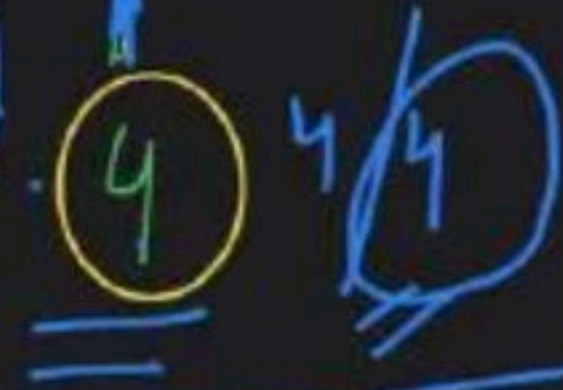
low update

Bridge check

$$1 < 2$$

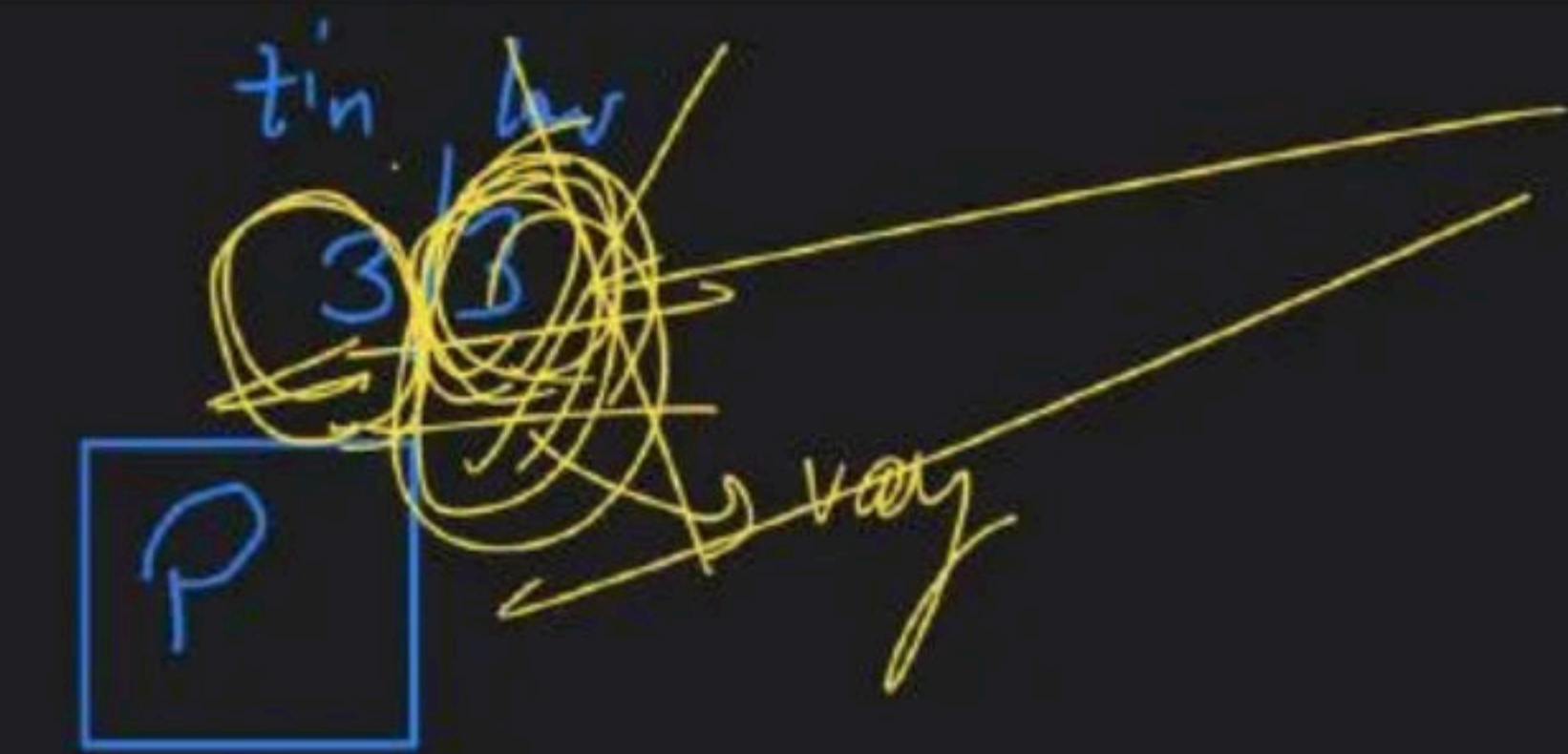
Bridge X

Bridge check



$$low[child] > tin[par]$$

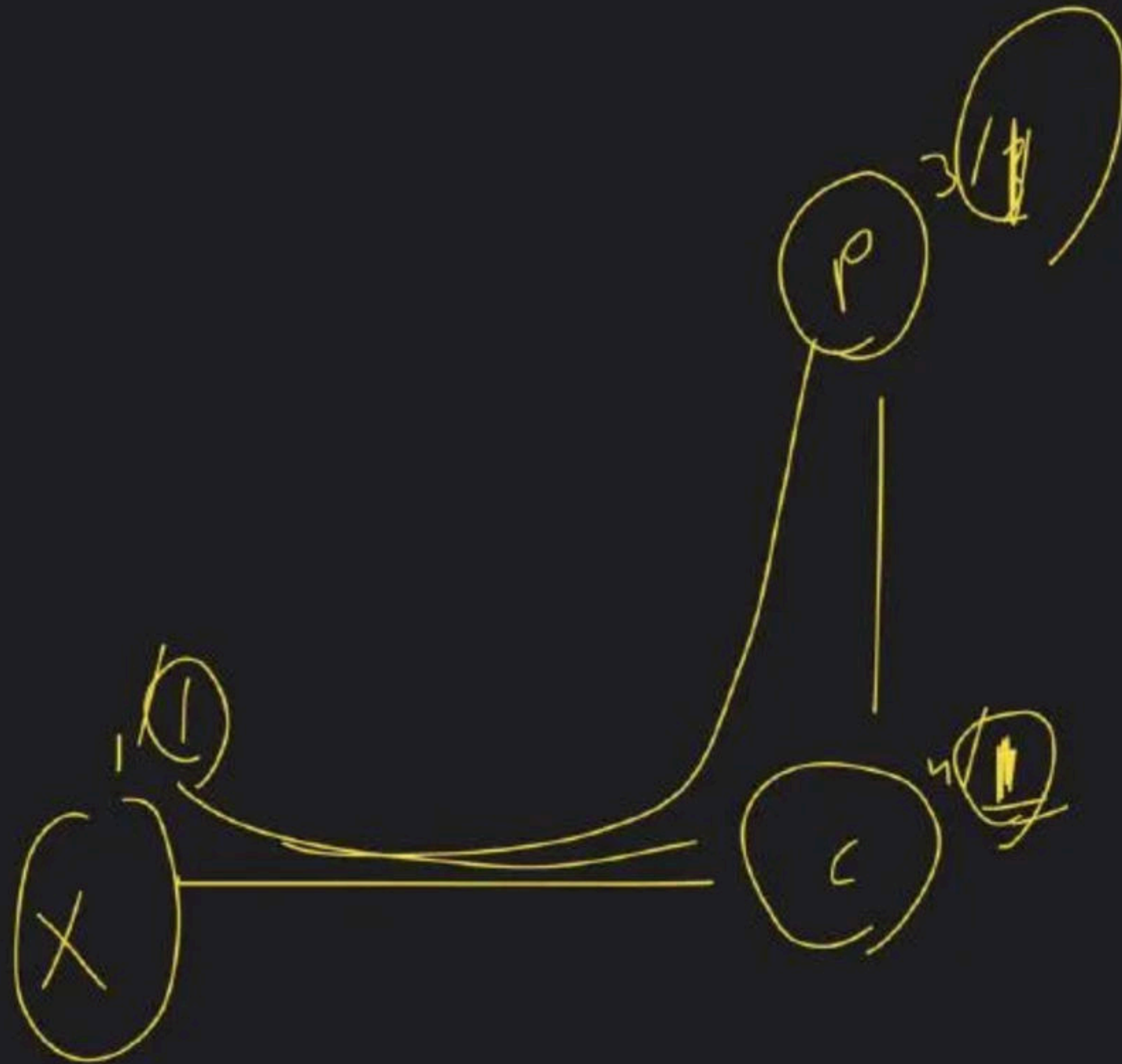
Bridge exist



1 < 2 → No Bridge

4 > 3 → Bridge

Yes



Contak → 18/25/27

Quations

→ skel + 9 pm

Graph Class - 7

Special class

① LL207

Course Schedule

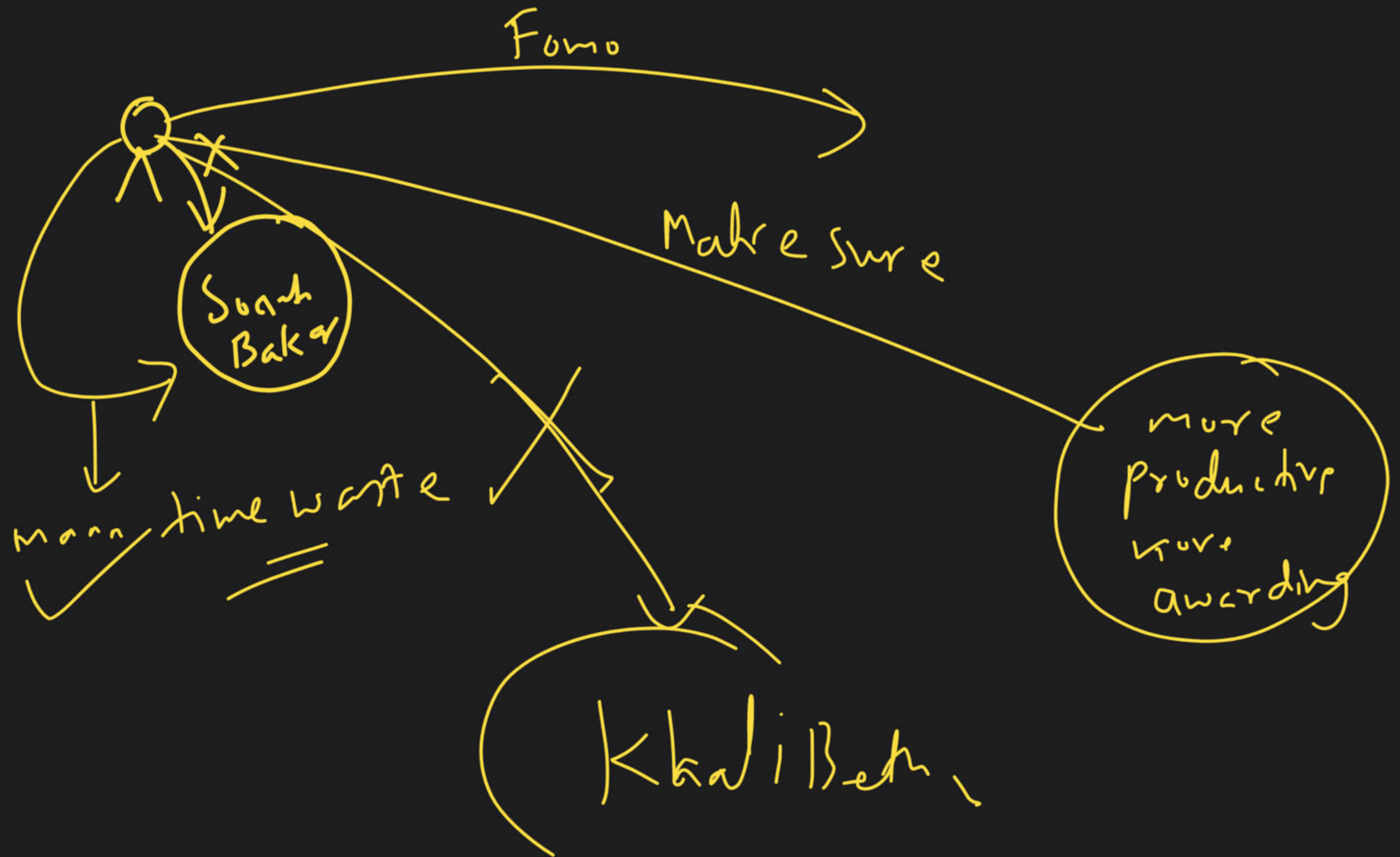
②

210

Course Schedule
↳ II

③

LL127



→ Word Ladder

hit → hot → dot → dog → cog

① Begin Word → hit

② End Word → cog

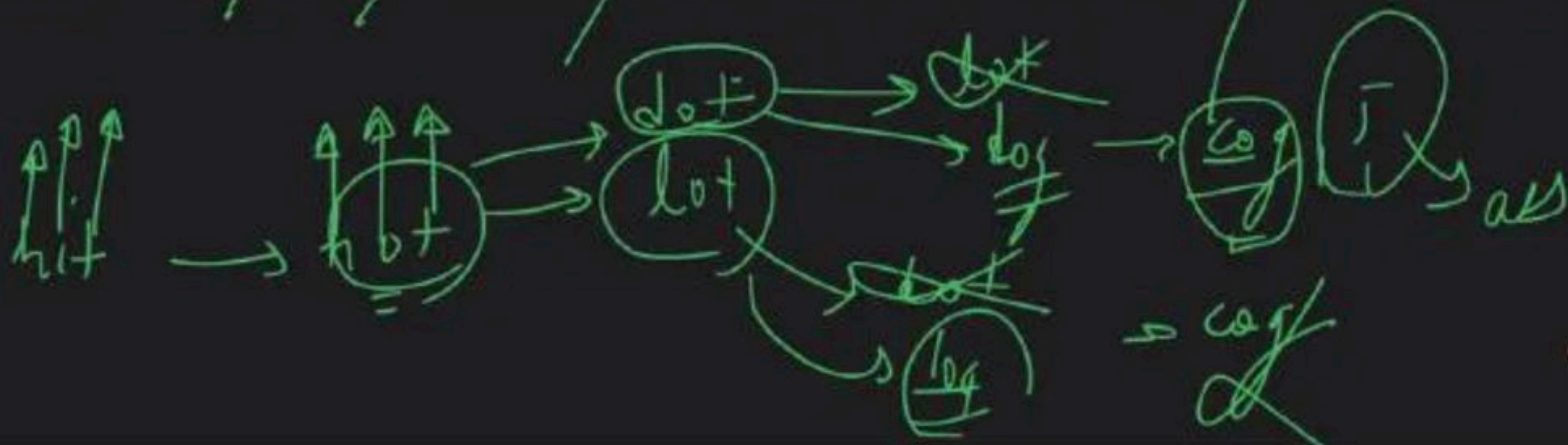
③ Dictionary → [hot, dot, dog, lot, log, cog]

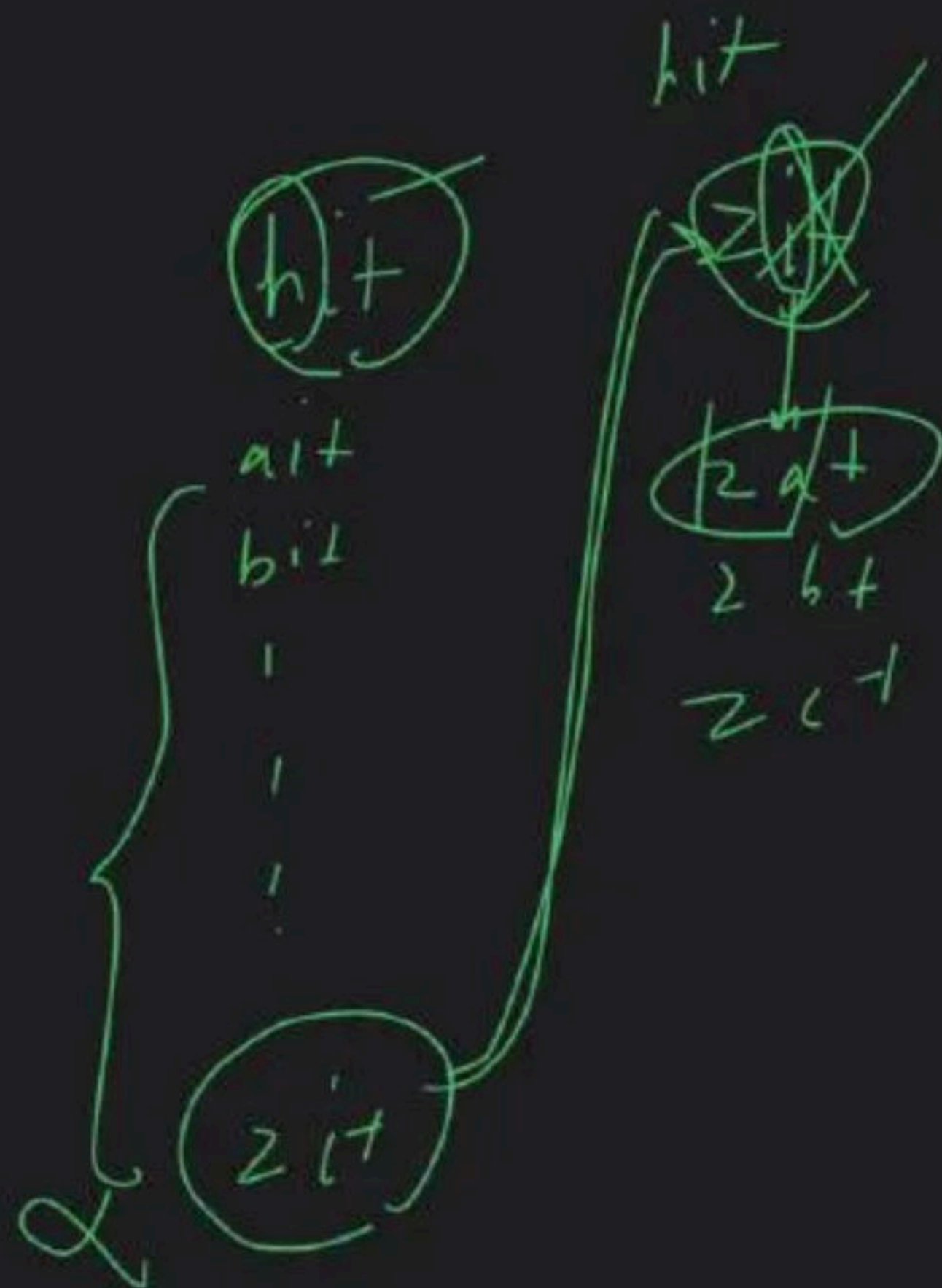
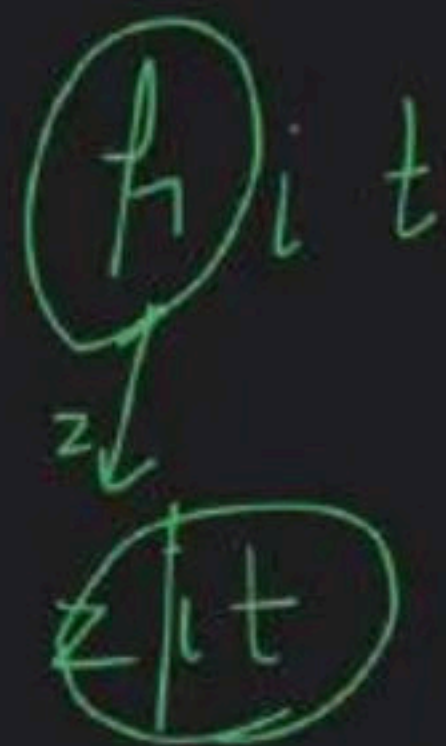
cog
log
lot
dog
dot
hot

Set (dictionary)

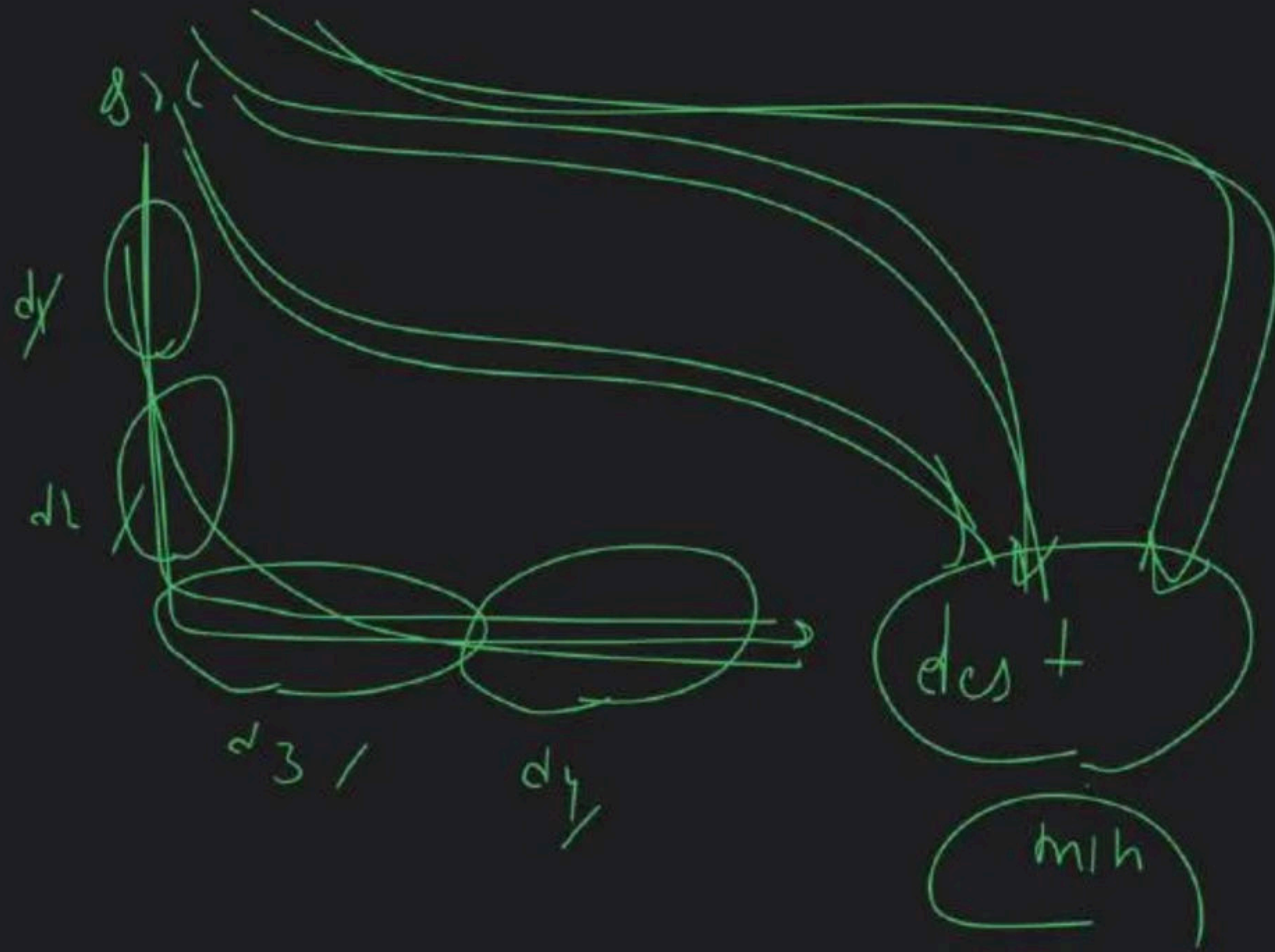
hit, 1	hot, 2	dot, 3	lot, 3	dog, 4	log, 4	cog, 5		
-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	-------------------	--	--

q < string, (h-l)

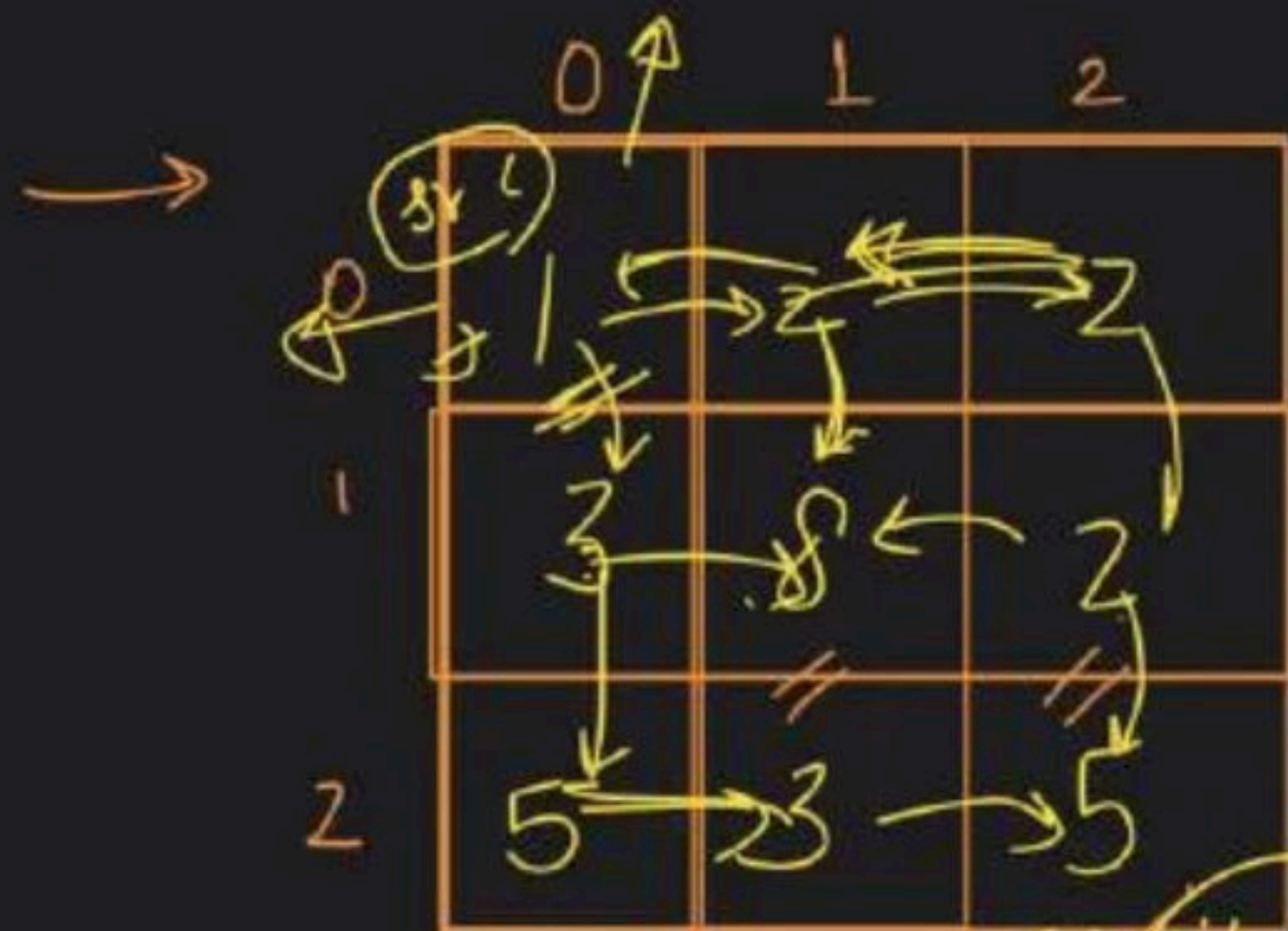




$\Rightarrow$ LC 1631, Path with Min $\sim$



$\leq$ effort $\sim$



dest
2, 2

dp

	0	1	2
0	0	1 0	0
1	2	5 0	0
2	2	2	2 2

dis

2 (2, 1)
3 (2, 0)
3 (2, 2)
0 (1, 2)
0 (0, 1)
2 (1, 1)
0 (0, 2)
2 (1, 0)
1 (0, 1)
0 (0, 0)

→ P2 < int, < int, int > >

< int, (int, int) >
↓
absolute
differe

diff = 0
n = 0
y = 0
X

diff = 1
n = 0
y = 1
X

diff = 0
n = 0
y = 2
X

diff = 2
n = 2, y = 2

→ Kad → gpm

↳ Bonus Session

→ Creedy